

THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY
Department of Electronic and Computer Engineering
ELEC 1100

Laboratory 3: Diodes and DC Regulation (5%)

A) Objectives:

- To study the diode characteristics.
- To construct a regulator to convert DC voltages.

B) Equipment:

- Diode (1N4148), Zener diode (1N4734), DC voltage regulator (LM7805).
- Resistors, capacitors.

C) Prelab (solution included)

- Q1 Indicate the polarity of the diodes below.

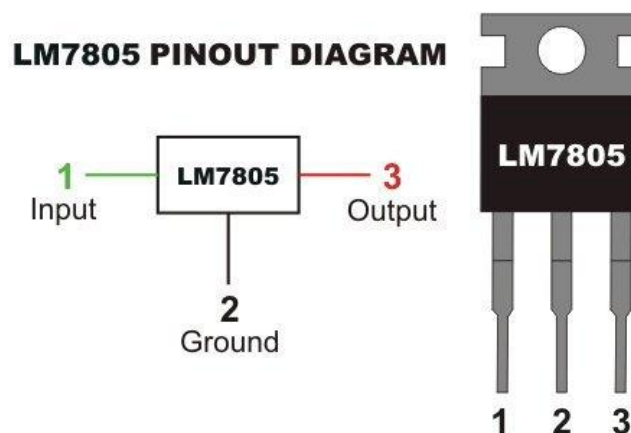


- Q2 What is the difference between a Zener diode and a diode?

Zener diode allows current to flow when the reverse voltage is larger than a certain value.

- Q3 How to identify each pin function of LM7805?

Let the number side of LM7805 face you, from its left to right are pin 1 (Input), 2 (Ground), and 3 (Output) as shown below.



D) Experiment Procedures:

- Review all the techniques in connecting components and measurement in the previous Lab1 and Lab2.
- Unless otherwise specified, all the negative terminals of equipment used in the experiment should connect to the GND  (the reference node, NOT the GND terminal of the real power supply) as given in the circuit diagram.

Experiment 1: Generating a 20V_{pp} Sine Wave (~10 mins)

(Note: Refer to Lab1, Experiment 6 if you forget the procedures.)

Step 1: Switch on the Function Generator and reset it by pressing  and then  .

Step 2: Set the function to **Sine Wave**, and the amplitude  to **10V**.

Step 3: Set the frequency to **1+(x/100)** kHz.

Note: In this lab, let x be the number represented by the 6th and 7th digits of your student ID. For example, if your student ID is 12345678, then x = 67.

Step 4: Turn on the DSO, select the language, and connect the DSO probe. Remember to set both the DSO setting and DSO probe to **1X**.

Step 5: Connect the Generator probe to the DSO probe using wires (positive to positive while negative to negative). You should see a **Sine Wave** now displaying on the screen with readings Pk-Pk=20V (20V_{pp}) and Freq=**1+(x/100)** kHz on the DSO.

****** TA Check 1: Show your generated sine wave to your TA. Each member of a group should demo once, using your own student ID for x.**

Quick Note

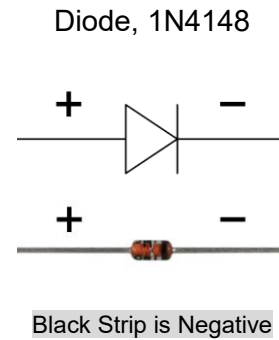
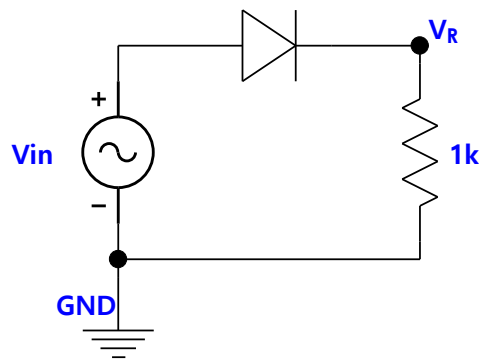
From our given components, the size of a diode is smaller than a Zener diode.



Experiment 2: Diode Circuit (~20 mins)

In this experiment, we study the diode circuit. A regular diode allows current to flow in one direction only: from its positive terminal to the negative.

Step 1: Find the regular diode 1N4148 (the **thin** one) and construct below circuit on the breadboard. **Notice the polarity of the diode.**



Step 2: Use the generated 20V_{pp} sine wave from **Experiment 1** as V_{in} .

Step 3: Connect CH1 of DSO to measure V_{in} and CH2 to measure V_R (voltage across the 1 k Ω resistor). Remember to connect both DSO probe negatives to GND (function generator negative terminal).

Step 4: Press the AUTO SET button to display both of the waveforms.

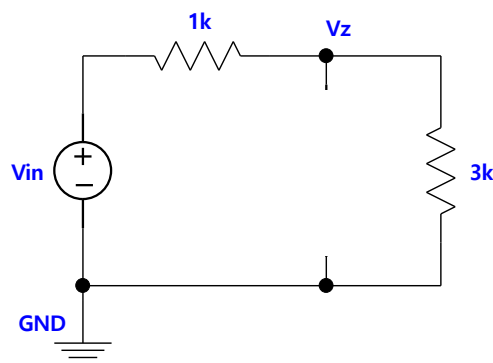
**** **TA Check 2: Show your measured waveforms to your TA.**

Q1: Sketch what you see on the DSO (both CH1 & CH2 waves, including the voltage & time scales and zero references displayed on the screen).

Q2: What is the difference between the two waveforms? Briefly explain your answer (Explain the feature of a diode).

Experiment 3: Zener Diode in Voltage Regulation (~30 mins)

Step 1: In this experiment, you will construct two circuits on one breadboard for comparison. Construct the below circuit on the top of the breadboard for saving space to have another one at the bottom.



Step 2: Use DC power supply to give V_{in} . Set V_{in} to 1V for now.

Step 3: Use the multimeter "V" mode to measure the voltage V_Z (versus GND) and record the reading in the summary sheet.

Step 4: Slowly increase V_{in} by 1V. Measure V_Z and record the reading.

Step 5: Repeat step 4 until $V_{in} = 16V$.

Q3: Complete the table of experiment results (2-resistor circuit) in the summary sheet.

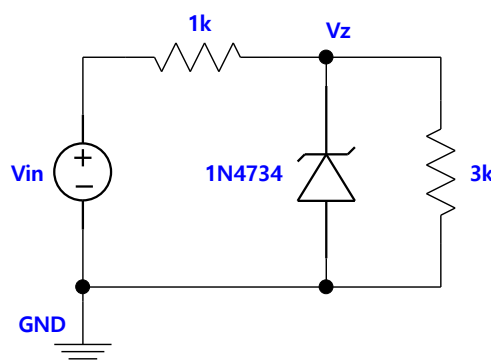
Notice that the two resistors in above circuit work as voltage divider. Therefore you should see from your record in the table that V_Z (voltage across the $3k\Omega$ resistor) is increasing linearly along with an increasing V_{in} . This obeys **Ohm's law**: $V_Z = \frac{3}{4}V_{in}$. In the end, when $V_{in} = 16V$, you should have $V_Z = 12V$.

Now we study the **Zener diode regulator** circuit.

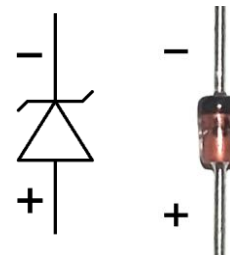
Zener diode is a special kind of diode that allows current to flow in the forward direction as normal (from diode positive to negative), but will also allow current to flow in the reverse direction when the voltage is above the "breakdown" value.

The voltage across the Zener diode in the breakdown region is almost constant. This ability of the Zener diode to control itself can be used to regulate or stabilize a voltage source against supply variation.

Step 6: Find the Zener diode 1N4734 (the **fat** one) and construct the below circuit at the bottom part of the breadboard. The power supply you already connect to the breadboard should also be the power source for the Zener diode circuit. **Notice the polarity of the Zener diode, which is REVERSELY connected in parallel with the $3k\Omega$ resistor.**



Zener Diode, 1N4734



Step 7: Set V_{in} from DC power supply to 1V.

Step 8: Use the digital multimeter to measure V_Z in the Zener diode circuit and record the reading in the summary sheet.

Step 9: Slowly increase V_{in} by 1V. Measure V_Z and record the reading.

Step 10: Repeat step 9 until $V_{in} = 16V$.

Q4: Complete the table of experiment results (Zener diode circuit) in the summary sheet.

**** **TA Check 3: Show your TA the measurement of both V_Z (in the 2-resistor circuit and the Zener diode circuit, respectively) when given $V_{in} = 16V$.**

Q5: Use the data you record in Q3 and Q4 to make a **line chart** of V_Z against V_{in} and fill in the summary sheet.

Below instructions are for your reference. Drawing by hand is also acceptable.

<https://www.edrawmax.com/line-graph/how-to-make-a-line-graph-in-word/>

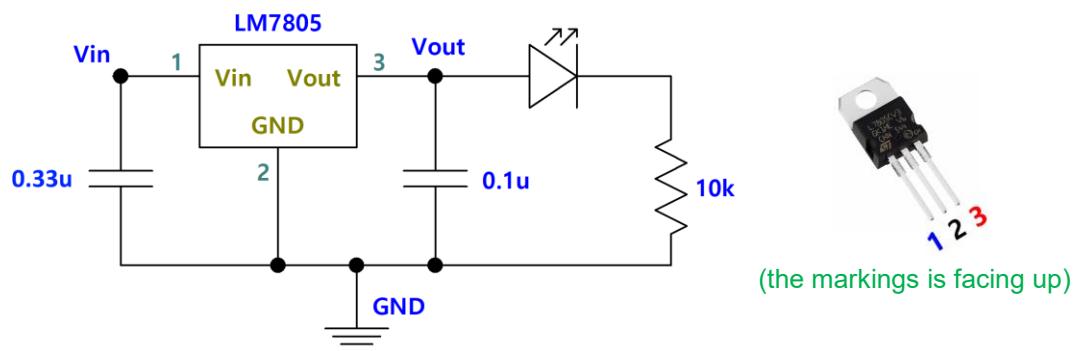
Notice that the Zener diode is working as a **voltage regulator**. Therefore you should see from the **line chart** that, unlike what you got in Q3, V_Z in Q4 is now stabilized by the Zener diode when V_{in} is large enough.

Q6: From your record in Q4, determine the breakdown voltage (give a rough number) of the Zener diode.

Experiment 4: LM7805 Voltage Regulator (~30 mins)

Now we start to construct the voltage regulator circuit for your project using the IC (Integrated circuit) LM7805.

Step 1: Construct below circuit on the breadboard. (You may refer to Tutorial notes for the breadboard arrangement).



The number on the capacitor is read as:

- Digits 1, 2: Significant bits
- Digit 3: Multiplier, i.e. 10^n
- Basic unit: pF ($p = 10^{-12}$)

E.g. "104" = 10×10^4 pF = **0.1 μ F**



Step 2: Set the power supply to 0V for now to give V_{in} .

Step 3: Set the multimeter to "V" mode to measure the voltage at V_{out} .

Step 4: Slowly increase the V_{in} (i.e. power supply) to about 8V, the maximum reading of the multimeter should be about 5V.

If the maximum reading is still close to 8V, turn off the power supply and check your circuit connection, and repeat Step 4. If the problem still exists, ask your TA for help.

CAUTION

Do NOT touch the regulator LM7805 as it gets hot when operating.

However, if a bad smell comes out, **turn off the power supply immediately and ask TAs for help.**

Step 5: Increase V_{in} to 12V and check if V_{out} is still about 5V. You should see the LED light on and a 5V reading on the multimeter.

**** TA Check 4: Show your TA that you have finished the circuit and obtained 5V output from 12V.

Q7: What is the multimeter reading (V_{out}) **displayed** when $V_{in} = 8V$?

Q8: What is the multimeter reading (V_{out}) **displayed** when $V_{in} = 12V$?

Q9: Use the results in Q7 and Q8, determine the line regulation of your LM7805 voltage regulator (Show your calculation).

Line regulation (You may refer to the Lecture 5 notes, pages 18-19) is to measure how stable the output voltage is with respect to the change of the input voltage, which given as:

$$\text{Line regulation} = \frac{\Delta V_O}{\Delta V_I}$$

After you have done Experiment 4, remove all of the components from your breadboard.

Remember to clean up your bench and return all the components, breadboard, tools, and cable box! A messy table will cost 1 point!